

Deployment & Pilot Runbook

ใช้ runbook นี้เป็นลำดับงานสำหรับ IT/Ops ก่อนเปิด pilot

1. Pre-Deployment

Step	Command/Evidence	Owner	Result
Confirm target commit	<code>git rev-parse HEAD</code>		
Install dependencies	<code>npm ci</code>		
Type check	<code>npx tsc --noEmit</code>		
Unit/migration tests	<code>npm test</code>		
Storage config	<code>npm run storage:check</code>		
Production readiness strict	<code>PRODUCTION_READINESS_STRICT=true npm run production:check</code>		

2. Required Environment

Variable	Purpose	Required before pilot
<code>DEPLOYMENT_PROFILE</code>	deployment mode such as <code>pilot</code>	yes
<code>DATABASE_URL</code>	Postgres connection	yes
<code>API_TOKEN</code>	bearer token for API	yes
<code>AUTH_SESSION_SECRET</code>	signed session secret	yes
<code>AUTH_LOGIN_CODE</code>	pilot login code policy	yes
<code>FILE_STORAGE_DIR</code>	persistent private file storage	yes
<code>FILE_STORAGE_MAX_UPLOAD_BYTES</code>	upload size policy	recommended
<code>FILE_STORAGE_ALLOWED_MIME_TYPES</code>	MIME allowlist	recommended

3. Database

1. Take backup before migration.
2. Apply migrations in order.
3. Run DB integration checks if available.
4. Record migration version and timestamp.
5. Keep rollback backup reference.

Evidence:

Backup ID:
Migration timestamp:
Operator:
Result:

4. File Storage

1. Configure persistent private storage.
2. Run `npm run storage:check`.
3. Upload/download a sample file in non-production test patient if allowed.
4. Confirm backup includes file assets.

5. Smoke Tests

Recommended:

```
npm run api:smoke
npm run frontend:workflow-smoke
npm run browser:workflow-smoke
npm run browser:api-workflow-smoke
```

If browser or Docker is not available, record why and run manual checklist.

6. Operations Drills

Drill	Evidence required	Result
Backup/restore	restored DB and file assets verified	
Rollback	previous build/database backup path verified	
Security incident tabletop	owner/action timeline recorded	
Printer failure drill	fallback/retry/report verified	

7. Pilot Day Checklist

- ☐ Owners online: clinic, clinical, pharmacy, billing, IT
- ☐ Monitoring owner assigned
- ☐ Backup owner assigned
- ☐ Incident owner assigned
- ☐ Rollback owner assigned
- ☐ Current backup completed
- ☐ Manual fallback forms/process ready
- ☐ Pilot defect tracker open
- ☐ Go/no-go decision recorded

8. Rollback Criteria

Rollback or pause pilot if:

- patient record cannot be opened or updated
- clinical note/signing workflow blocks care
- billing totals are materially wrong
- pharmacy stock or controlled-drug audit is wrong
- auth/privacy incident occurs
- backup/restore path cannot be confirmed
- printer/scanner issue blocks pharmacy and fallback fails

9. Post-Pilot Review

Item	Result
defect count by severity	
user feedback summary	
financial/pharmacy reconciliation completed	
audit log spot check completed	
next pilot decision	